

Financial Sales and Profitability Analysis

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn

Dataset: Financial Sales Dataset

Analysis Type: Exploratory Data Analysis and Business
Insight Generation

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Date: 30 Sep 2026

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1. Project Overview

This project analyzes a financial sales dataset using **Python and Pandas** to identify key revenue drivers, profitable customer segments, country-wise performance, product performance, and the impact of discounting on profitability.

The main objective was to transform raw business data into actionable insights that can support pricing, discounting, product, and customer-segment decisions.

2. Project Objectives

The objective of this project was to analyze financial sales data using Python and Pandas. The analysis evaluates sales, profit, customer segments, countries, products, discount bands, and segment-product combinations.

The project aims to identify the major revenue and profit drivers, understand the impact of discounts on profitability, identify low-performing areas, and provide data-driven business recommendations.

3. Dataset Overview

The dataset contains 700 records and 12 variables related to financial sales performance.

The main fields include Segment, Country, Product, Discount Band, Units Sold, Manufacturing Price, Sale Price, Gross Sales, Discounts, Net Sales, COGS, and Profit. 12) flags an issue.

Dataset attribute	Details
Records	700
Columns	12
Analysis tool	Python and Pandas

Dataset attribute	Details
Visualization tools	Matplotlib and Seaborn
Main focus	Sales, profit, margins, discounts, and loss-making combinations

4. Data Preparation

The dataset was loaded from a CSV file using Pandas. Data quality checks were performed to inspect column names, data types, missing values, and duplicates.

The column names were standardized by removing extra spaces and converting them into lowercase snake_case. Blank values in the Discount Band field were checked against the Discounts field. These records had a discount amount of zero, so the blank discount categories were correctly labelled as No Discount.

Additional metrics were created, including Profit Margin %, Discount Rate %, and Revenue per Unit.

- Sales=Gross Sales–Discounts
- Profit= Sales–COGS
- Profit Margin %= $\frac{\text{Net Sales}}{\text{Profit}} \times 100$

5. Key Findings

Analysis area	Key finding
Customer segments	Government generated the highest total profit: ₹11.39M
Segment efficiency	Channel Partners had the highest profit margin: 73.13%
Segment risk	Enterprise recorded a loss of about ₹0.61M with a margin of -3.13%

Analysis area	Key finding
Country sales	USA had the highest total sales: ₹25.02M
Country profit	France had the highest total profit: ₹3.70M
Country efficiency	Germany had the highest profit margin: 15.66%
Top product	Paseo had the highest sales: ₹33M and profit: ₹4.7M
Product efficiency	Amarilla had the highest profit margin: 15.86%
Discount sales	Medium discount generated the highest sales: ₹38.78M
Discount profit	Low discount generated the highest profit: ₹6.10M
Best discount margin	No Discount had the highest margin: 21.86%
Weakest discount margin	High discount had the lowest margin: 9.07%
Best combination	Government–Paseo generated ₹3.0M profit from ₹14.8M sales
Key risk combination	Enterprise–Carretera had an overall loss of ₹2.2M

6. Power BI Dashboard and Visualizations

❖ Dashboard Overview

A Power BI dashboard was developed to present the key findings from the Pandas analysis in an interactive and business-friendly format. The dashboard enables users to monitor overall sales and profitability, compare performance across segments, countries, products, and discount bands, and identify areas requiring business attention.

The dashboard acts as the visualization layer of this project, while Python and Pandas were used for data cleaning, calculation of metrics, exploratory analysis, and business insight generation.

❖ Dashboard KPIs

The dashboard includes KPI cards to summarize overall business performance. These KPIs include

- Total Sales

- Total Profit.

❖ Dashboard Visuals

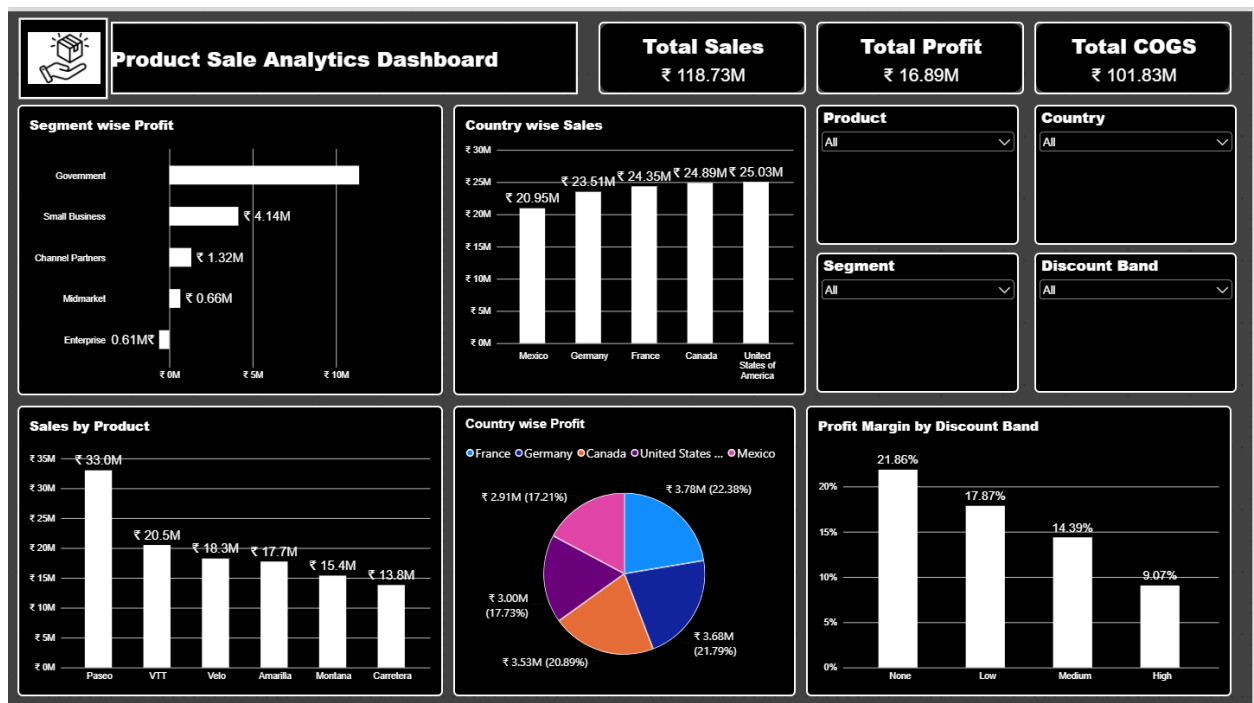
Visual	Purpose
KPI Cards	Show overall sales, profit and COGS
Profit by Segment	Identify the most and least profitable customer segments
Sales by Country	Compare revenue contribution across countries
Profit by Country	Compare profit contribution across countries
Profit by Product	Identify products that drive or weaken profitability
Profit Margin by Discount Band	Measure the effect of discounting on profitability
Segment–Product Performance	Identify top-performing and loss-making combinations
Slicers	Filter results by Country, Segment, Product, and Discount Band

❖ Dashboard Insights

- The Power BI dashboard visually confirmed the results of the Pandas analysis. Government was the highest-profit customer segment, generating approximately ₹11.39M in profit, while Channel Partners had the highest segment profit margin at 73.13%.
- The USA generated the highest net sales at approximately ₹25.02M, but France generated the highest total profit at approximately ₹3.70M. Paseo was the leading product by total sales and total profit, while Amarilla achieved the highest product-level profit margin.

- The discount analysis showed that No Discount transactions had the highest profit margin at 21.86%, whereas High discount transactions had the lowest margin at 9.07%. This visualization confirmed that higher discounts were associated with weaker profitability.
- The dashboard also highlighted Enterprise–Carretera as a major loss-making combination, with an overall loss of approximately ₹2.2M.

❖ Dashboard Screenshot



7. Enterprise-Carretera Combination Deep Dive

the **Enterprise–Carretera** combination was the least profitable and recorded a net loss of approximately ₹2.2M. This loss is likely a major driver of Enterprise's overall negative profitability and should be investigated for unfavorable pricing, high COGS, excessive discounting, or an unprofitable product–segment fit.

- The Enterprise–Carretera combination was identified as the highest-risk area in the analysis. It generated an overall net loss of approximately ₹2.2M.
- Out of 15 Enterprise–Carretera transactions, 12 transactions were loss-making and only 3 were profitable. Therefore, 80% of transactions under this combination generated losses.
- The highest single-transaction profit was ₹5,304.38, while the largest individual loss was ₹38,046.25. This indicates that the current selling price, discount policy, and COGS structure were not sufficient to reliably cover costs.

Metric	Result
Total transactions	15
Loss-making transactions	12
Profitable transactions	3
Loss-making transaction rate	80%
Largest transaction profit	₹5,304.38
Largest transaction loss	-₹38,046.25
Overall profit/loss	-₹2.2M

8. Recommendations

1. Protect and expand Government–Paseo because it was the largest profit-generating segment-product combination.
2. Review Enterprise–Carretera immediately. The company should reassess its pricing, discount levels, COGS, customer contracts, and product positioning in the Enterprise segment.
3. Avoid broad High discount strategies. High discounts had the lowest profit margin of 9.07%, while No Discount transactions achieved the highest margin of 21.86%.

4. Prefer targeted Low-to-Medium discount strategies where discounts are necessary. Medium discounts supported the highest sales, while Low discounts generated the highest total profit.
 5. Study Channel Partners and Amarilla to identify successful pricing, cost-control, and product-mix practices that can improve low-margin areas.
 6. Investigate USA-level sales because it had the highest revenue but the lowest country-level profit margin., and at comparable ages newer cohorts track older ones closely.
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9. Conclusion

This project demonstrated how Pandas can be used to analyze a business sales dataset, calculate profitability metrics, perform grouped analysis, identify patterns, create visualizations, and generate business recommendations.

The analysis showed that high sales alone do not guarantee high profitability. Profitability was strongly influenced by customer segment, product selection, country, cost structure, and discounting strategy. The most important action area is the Enterprise–Carretera combination, where frequent losses indicate an unsustainable strategy.